



HAI: Human-AI Interaction

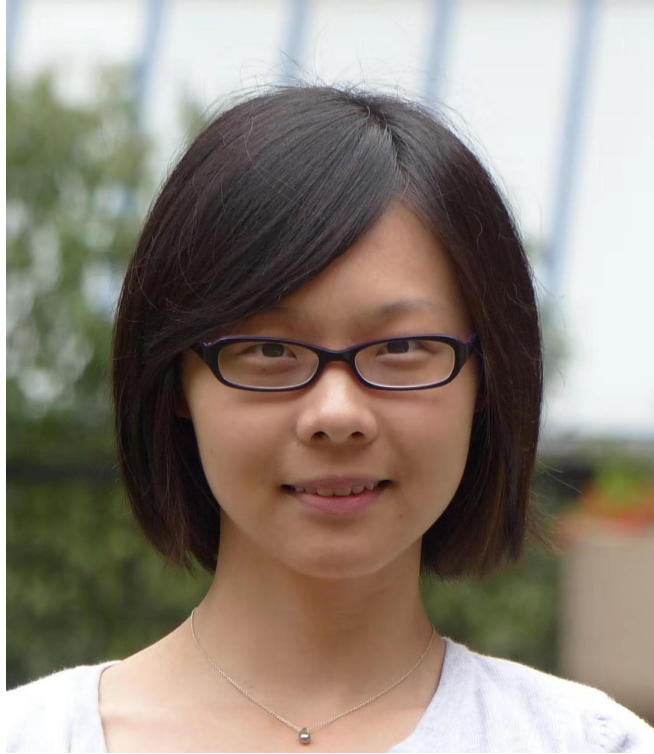
Fall 2025

Haiyi Zhu & Motahhare Eslami
Human Computer Interaction Institute (HCII)
School of Computer Science (SCS)

Today's class

- Introduction of the instructor team
- Why we're doing this course
- Course schedule & logistics
- Discussion: What benefits and harms do you perceive in using generative AI tools (e.g., ChatGPT) in this class? Why?

Instructors



Haiyi Zhu
Associate Professor



Motahhare Eslami
Assistant Professor

Haiyi's Research

AI Technologies



Social Systems

Groups
Communities
Organizations
Market
Society

The integration of AI technologies and algorithms into various aspects of social groups, communities and societies to **enhance, automate, or optimize social processes, interactions, and decision-making.**

AI-Supported Large-Scale Social Systems



AI-Supported Public Services

Child Welfare, Housing Services, Building Services



AI-Supported Online Content Moderation

Wikipedia, Reddit



Empowering and Enhancing Gig Workers

Gig platforms (e.g., Uber, Lyft, Upwork, Instacart)



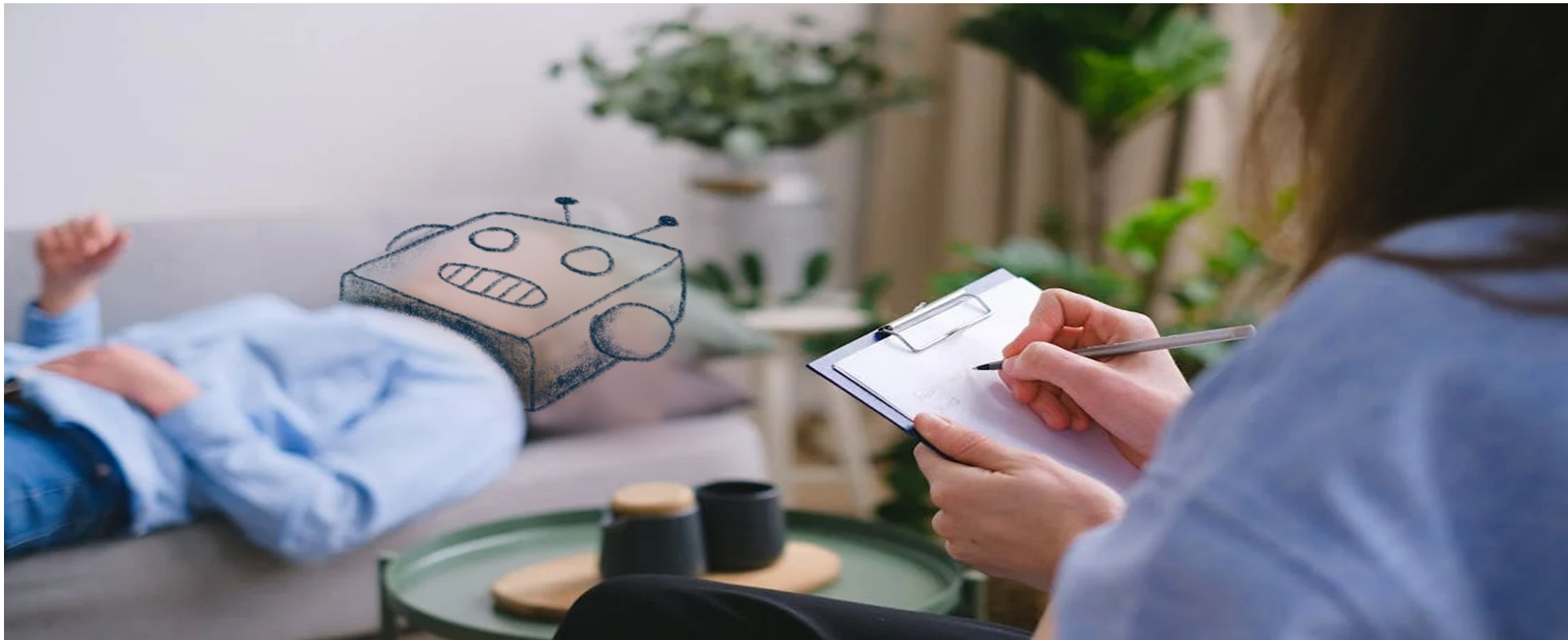
AI-Augmented Mental Health Support

With Robert Kraut and Sherry Wu

We are recruiting Research Assistants!

Project: Building Virtual Patients to Deliver Experiential Training for Mental Health Providers

Email me <haiyiz@andrew.cmu.edu> if you are interested!



Motahhare's research is broadly in human-computer interaction, social computing, and data mining, with a focus on “Responsible AI”



MIT Technology Review

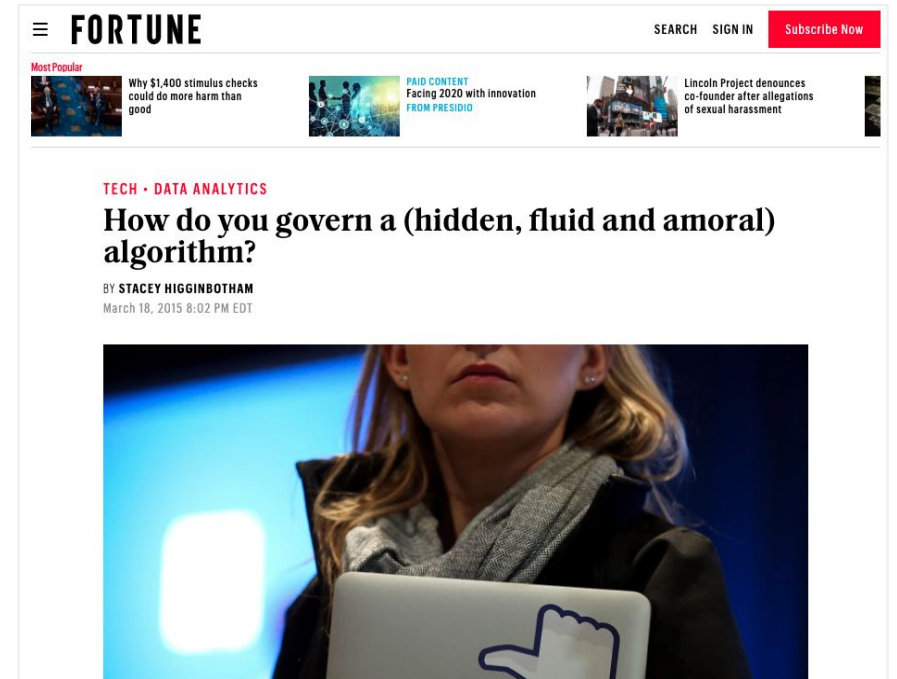
Algorithm Awareness

How the news feed on Facebook decides what you get to see.

by **Karrie Karahalios**

October 21, 2014

Increasingly, it is algorithms that choose which products to recommend to us and algorithms that decide whether we should receive a new credit card. But these algorithms are buried outside our perception. How does one begin to make sense of these mysterious hidden forces?



Motahhare's Research

Empowering **everyday users and impacted stakeholders** of AI systems, especially **those whose voices are often unheard**, to have a say in the design, development, and evaluation of accountable systems



4A Lab: AI Accountability, Awareness and Action Lab



Create a future with fair technology for everyone with everyone

We harness the power of the crowds, specifically you, to improve machine learning and algorithmic fairness. Create an account to join our community, where you can audit biases and participate in discussions!

[See All Projects](#)



W. H. Deng, C. Wang, H. Han, J. Hong, K. Holstein*, M. Eslami*. *WeAudit: Scaffolding User Auditors and AI Practitioners in Auditing Generative AI*. Proceedings of ACM in Human Computer Interaction (CSCW), 2025.

Supported by **NSF + Amazon** Fairness in AI (FA) Program, **Google** Research Scholar Program, **Cisco** Ethics in AI, **Microsoft Research** AI & Societyity Fellowship, and **Notre Dame-IBM** Tech Ethics lab

School of Computer Science

TA Team



Shivani Kapania (Section A)
PhD Student



Lisa Egede (Section B)
PhD Student

Shivani's Research

My research focuses on **cultures of safety** and **labor impacts** within (and resulting from) the AI supply chain, explored through two complementary lines of inquiry:

- 'Upstream', for data workers
- 'Downstream', within the autonomous vehicle industry



Lisa's Research

My research aims to understand Black people's experiences using and designing AI system, as well as the broader socio-technical challenges faced by historically excluded communities.

- Understanding how racially minoritized communities are designing *culturally tailored* AI systems



Why we're doing this course

- Most AI/ML courses are disconnected from real-world problems

What is AI - Artificial Intelligence?

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Computers doing things that we expect people to be able to do

What is AI - Artificial Intelligence?

Computers doing things that we expect people to be able to do

- **Learning from examples:** machine learning, computer vision, natural language processing etc.
- **Rule-based AI/expert systems:** mimic an expert.

What do we mean by AI?

Computers doing things that we expect people to be able to do

- **Learning from examples:** machine learning, computer vision, natural language processing etc.
- **Rule-based AI/expert systems:** mimic an expert.

Machine Learning (ML)



"A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E ."

-- Tom M. Mitchell

My personal experience with learning ML

Theorem. Let $\mathcal{H}$ be given, and let $d = \text{VC}(\mathcal{H})$. Then with probability at least $1 - \delta$, we have that for all $h \in \mathcal{H}$,

$$|\varepsilon(h) - \hat{\varepsilon}(h)| \leq O \left(\sqrt{\frac{d}{m} \log \frac{m}{d} + \frac{1}{m} \log \frac{1}{\delta}} \right).$$

Thus, with probability at least $1 - \delta$, we also have that:

$$\varepsilon(\hat{h}) \leq \varepsilon(h^*) + O \left(\sqrt{\frac{d}{m} \log \frac{m}{d} + \frac{1}{m} \log \frac{1}{\delta}} \right).$$

I've used this beautiful mathematical result zero times for building interactive AI

AI affects many facets of human life & society

Twitter taught Microsoft's AI chatbot to be a racist asshole in less than a day

Amazon Pushes Facial Recognition to Police. Critics See Surveillance Risk.



AI affects many facets of human life & society

TECHNOLOGY

Sam Altman Says AI Agents Will Transform the Workforce in 2025 In a

new blog post, the famous OpenAI CEO reflected on his firing, what the company could do better, and a pursuit of 'superintelligence.' [🔗](#)

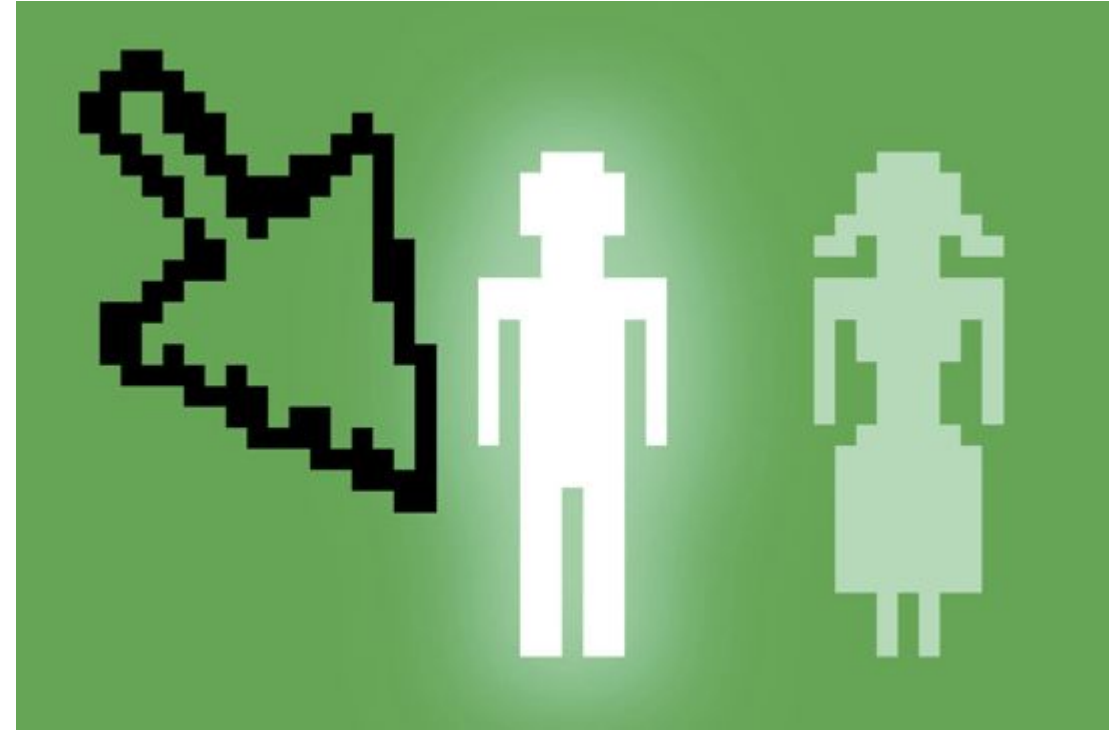
BY BEN SHERRY, STAFF REPORTER @BENLUCASSHERRY

JAN 6, 2025

Unintended biases



African-American defendants are substantially more likely than **White-American** defendants to be incorrectly classified as high risk by the algorithms.



Google Ads displayed more high paying jobs for **male users** than for **female users**.

Working with AI: Measuring the Occupational Implications of Generative AI

[Kiran Tomlinson](#), [Sonia Jaffe](#), Will Wang, [Scott Counts](#), [Siddharth Suri](#)

July 2025

[Download BibTex](#)

Microsoft researchers have revealed the 40 jobs most exposed to AI—and even teachers make the list

 BY PRESTON FORE
STAFF WRITER, EDUCATION

July 31, 2025 at 11:31 AM EDT



<https://www.microsoft.com/en-us/research/publication/working-with-ai-measuring-the-occupational-implications-of-generative-ai/>

Top 10 Jobs Safe From Generative AI

1. Phlebotomists
2. Nursing Assistants
3. Hazardous Materials Removal Workers
4. Helpers—Painters, Plasterers, etc.
5. Embalmers
6. Plant and System Operators, All Other
7. Oral and Maxillofacial Surgeons
8. Automotive Glass Installers and Repairers
9. Ship Engineers
10. Tire Repairers and Changers

40 Jobs Most at Risk from AI

1. Interpreters and Translators
2. Historians
3. Passenger Attendants
4. Sales Representatives of Services
5. Writers and Authors
6. Customer Service Representatives
7. CNC Tool Programmers
8. Telephone Operators
9. Ticket Agents and Travel Clerks
10. Broadcast Announcers and Radio DJs
11. Brokerage Clerks
12. Farm and Home Management Educators

Why we're doing this course

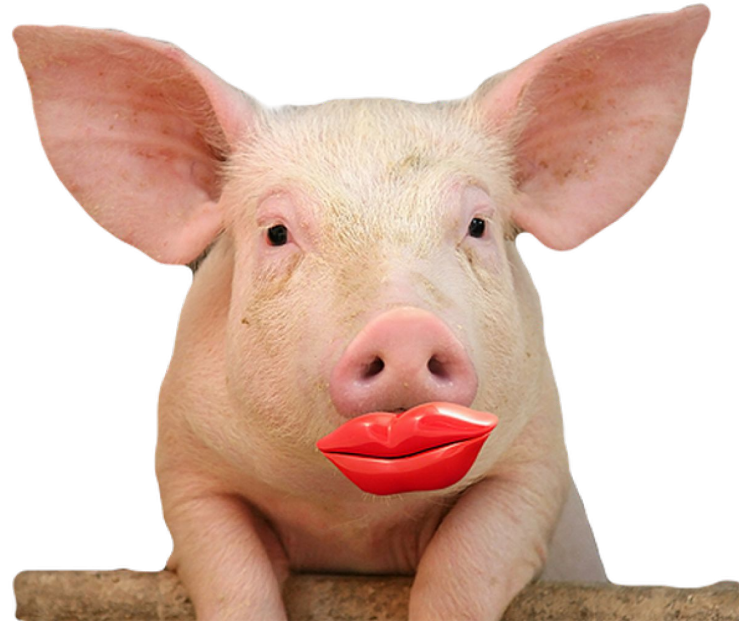
- Most AI/ML courses is disconnected from real-world problems
- **We want to teach AI/ML in a way that is grounded in real-world problems**

Why we're doing this course

- Most AI/ML courses consider “user-interfaces” or human factor as an afterthought; and focus narrowly on algorithms

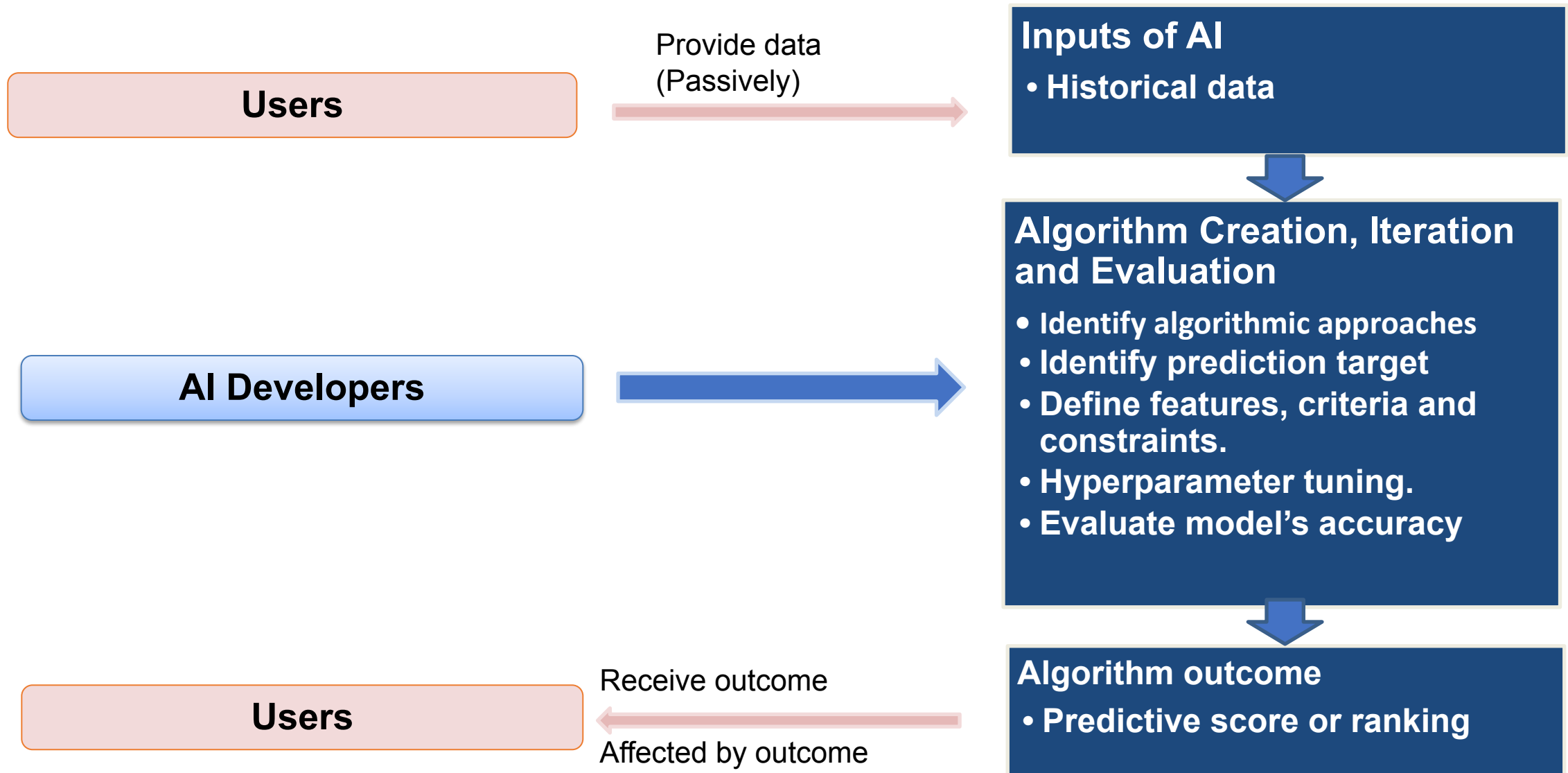
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The phrase to put "**lipstick on a pig**" means making superficial or cosmetic changes to a product in a futile effort to disguise its fundamental failings.

Traditional development pipeline



Andrew Ng: Building Faster with AI

Y COMBINATOR PRESENTS

AI STARTUP SCHOOL

Feedback loop to derisk customer acceptance

With rapid engineering, this becomes the bottleneck

Get Feedback (PM)

Build Software (Engineering)

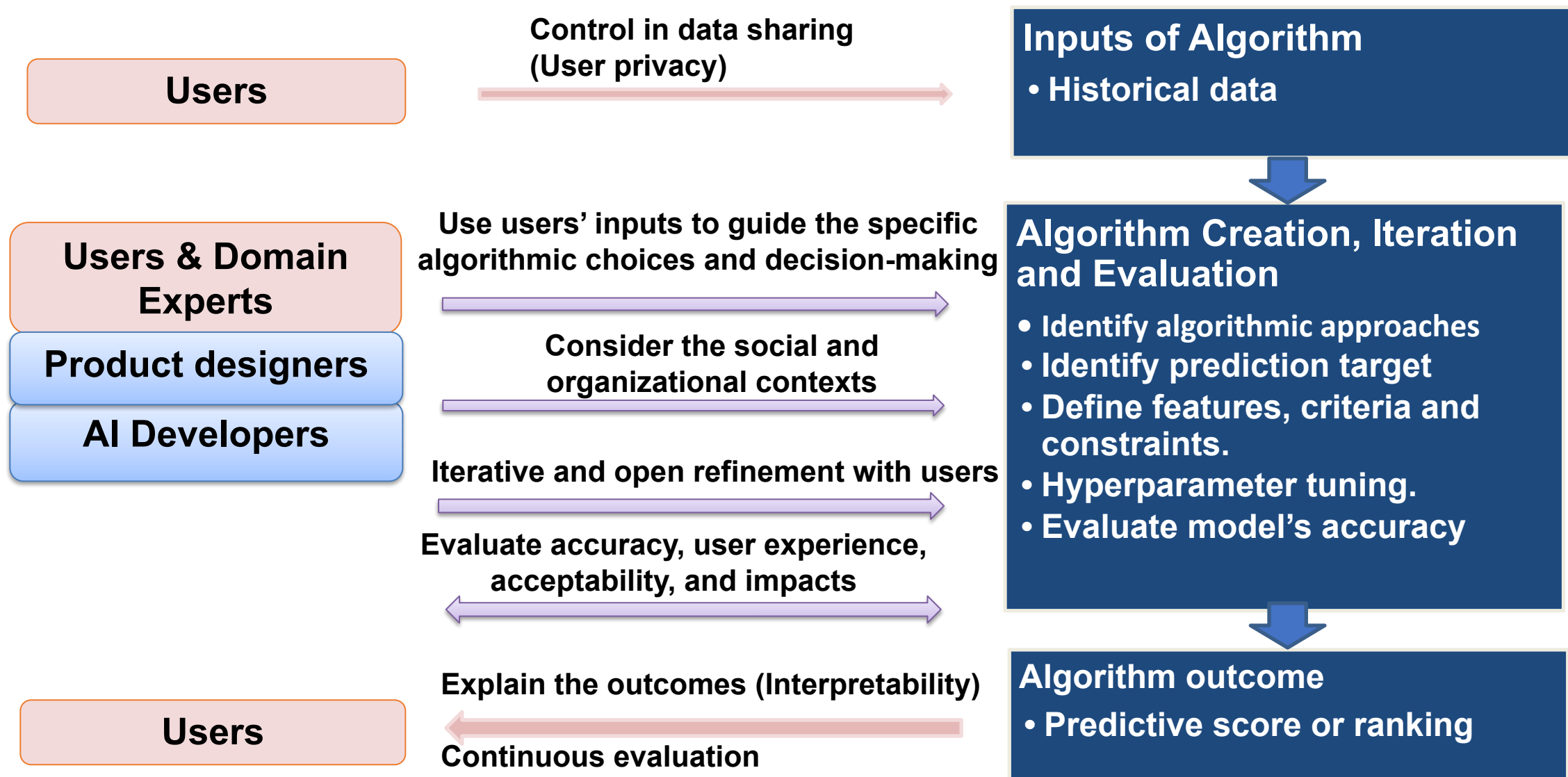
Prototype

- Changing Eng:PM ratios

Andrew Ng

17:28 / 43:57 • The Role of Product Management >

Human-centered approach



Why we're doing this course

- Most AI/ML courses are disconnected from real-world problems
- **We want to teach AI/ML in a way that is grounded in real-world problems**
- Most AI/ML courses consider “user-interfaces” or human impact as an afterthought; and focus narrowly on algorithms
- **We want to teach a new human-centered way to build AI systems**

Let's do it!

Our teaching philosophy

- Put people first!
- DO NOT learn machine learning from the scratch
 - Instead, we will use building blocks, and leverage existing tools and library!
- We'll help you think it through, but also “do it through” development
 - Nothing gives you a sense of possibilities and limitations like building it
- To learn something, teach it
 - True in general, but for AI, industry folks have all sorts of ideas (often wrong).
 - To succeed, you need to learn how to teach them

What should you expect to do in this course?

- **Thinking it through:** Readings, reflections, lectures
- **Doing it through:** Assignments and projects
- **Teaching it through:** In class discussions, group exercises, deliberation, and debate, etc.

What should you expect to do in this course?

- **Thinking it through:** Readings, reflections, lectures

⋮	▼ Module I: Introduction & Guidelines	✓ ▼	+	⋮
⋮	📄 Session 1: Introduction to Human-AI Interaction	✓		⋮
⋮	📄 Session 2: Practical Machine Learning	✓		⋮
⋮	📄 Session 3: Perspectives on Human-AI Interaction	✓		⋮
⋮	💬 Session 3 Reading Reflection Sep 3 1 pts	✓		⋮
⋮	📄 Session 4: Guidelines of HAIL	✓		⋮
⋮	💬 Session 4 Reading Reflection Sep 8 1 pts	✓		⋮

Activities and Grading

Reading Reflections (15%)

- Read the assigned article and submit a paragraph of reflection on Canvas discussion
- Deadlines: should be submitted before class
 - **Mons and Weds @ 12 pm NOON (not midnight!) for both sections**

Course trajectory

- **Module I:** Introduction & Guidelines
- **Module II:** Fairness, Accountability, Transparency, and Ethics (FATE) in Human-AI Interaction
- **Module III:** Capabilities and Limitations of Advanced ML
- **Module IV:** AI Governance and Society

[Schedule](#)

What should you expect to do in this course?

- **Thinking it through:** Readings, reflections, lectures
- **Doing it through:** Assignments and projects

Assignments

- **Assignment 1:** Train a machine learning model to determine if someone should get a mortgage, see how well it works
- **Assignment 2:** Use Fairlearn library to audit and improve your own model
- **Assignment 3:** Visualize data that goes into an AI system and explain AI system's behavior
- **Assignment 4:** Generative AI

Activities and Grading

Assignments (40%)

- 4 assignments
- You will each receive 5 free late days.
 - You can budget late days as you like, for instance 3 free days late for one assignment, 2 for another, or all 5 for one.
 - for students who do not use their free late days, they receive 0.5% extra credit for each day at the end of the semester.

Do you know enough programming?

You need to know some Python

Example tasks:

Python: Take data as a csv, and transpose all rows into columns

Project

- Individual-based project
- Multiple tracks
- [Project information](#)
- Final project (30%)

Project Schedule

- **Proposal** (Due 9/26)
 - One-two paragraphs
 - Which track do you want to choose
 - Your initial plan
- **Project update** (Due 10/31)
- **Feedback session** (11/12: section A, 11/13: section B)
- **In-class presentation** (last week of class: 12/01-12/04)
- **Final submissions** (Due 12/08)

What should you expect to do in this course?

- **Thinking it through:** Readings, reflections, lectures
- **Doing it through:** Assignments and projects
- **Teaching it through:** In class discussions, deliberation, exercises, and debate, etc.

Activities and Grading

Attendance and participation (15%)

- In-class activities (1-2 times a week)
- Small group discussions and/or activities in the classroom
- Posting the activity on the related Slack channel
- Each student will be excused for at most two missing sessions

Expectation

Attend the class in person

If you are sick, please stay home and communicate with us.

Syllabus

Communication

Canvas

- Schedule, Readings, Slides, Assignments and Projects

Slack

- Questions!
- Section A
 - [Join the slack](#)
- Section B
 - [Join the slack](#)

What questions do you have?



How can I help you today?

Brainstorm content ideas
for my new podcast on urban design

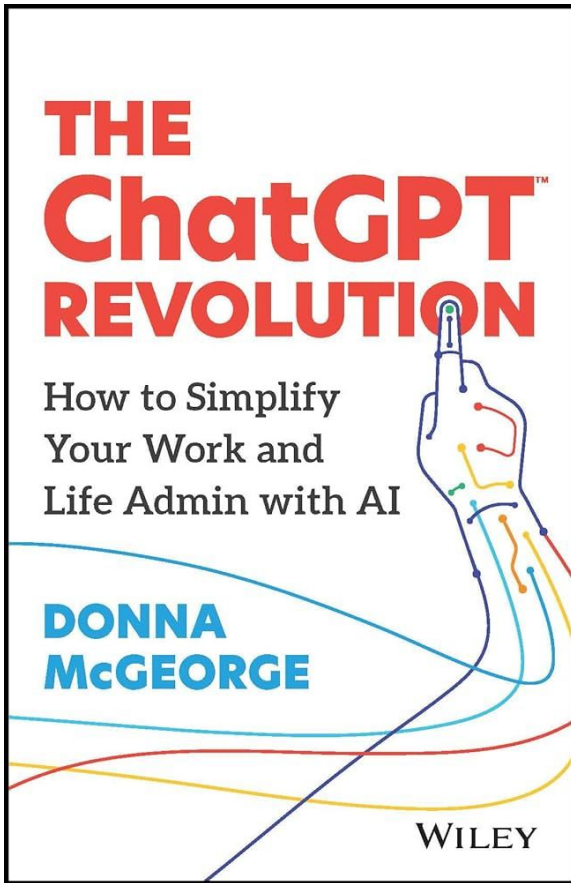
Compare storytelling techniques
in novels and in films

Create a workout plan
for resistance training

Recommend a dish
to bring to a potluck

 Message ChatGPT...





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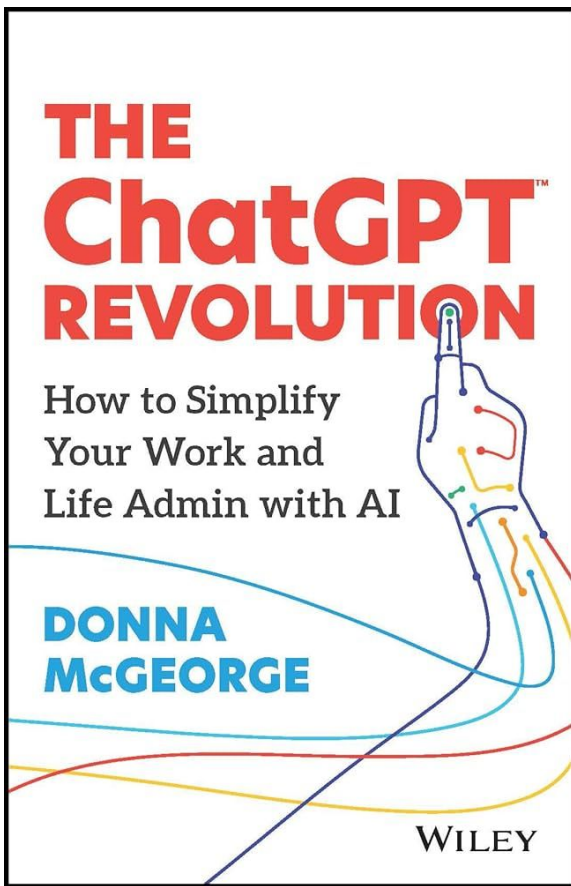
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Message ChatGPT...



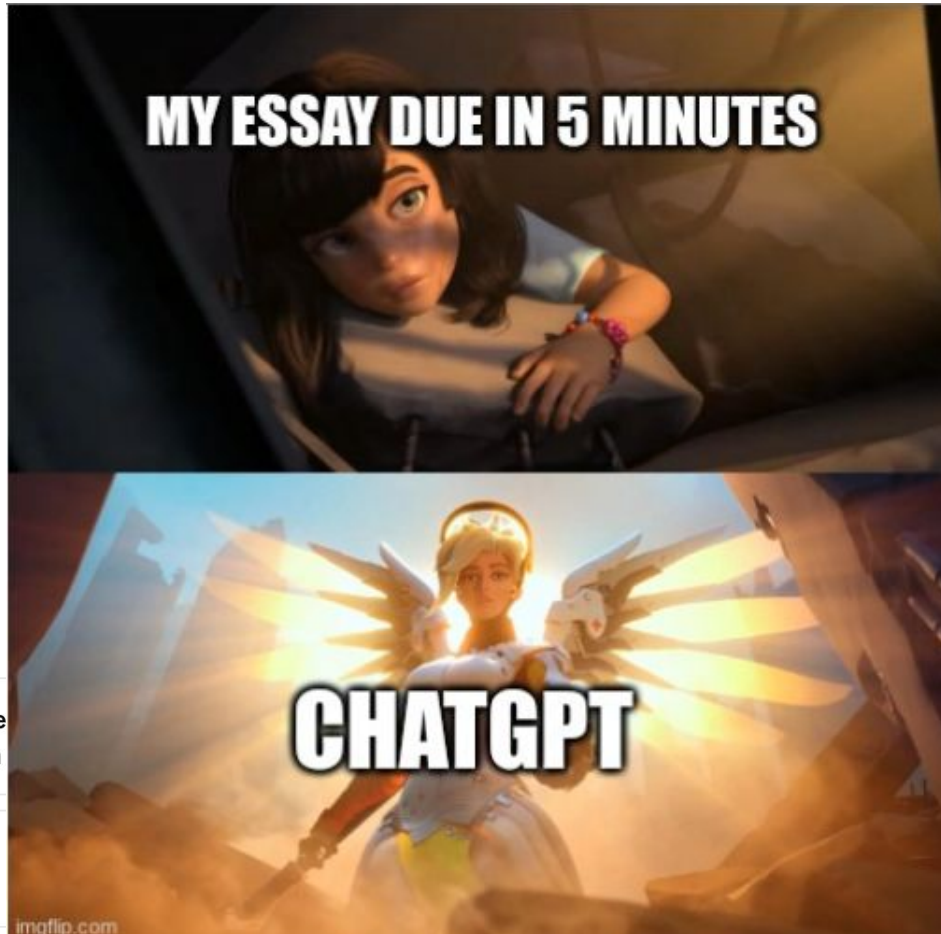


form content ide
new podcast on

Create a workout plan
for resistance training



Message ChatGPT...



THE ChatGPT[™] REVOLUTION

How to Simplify
Your Work and
Life Admin with AI

DONNA
McGEORGE

WILEY

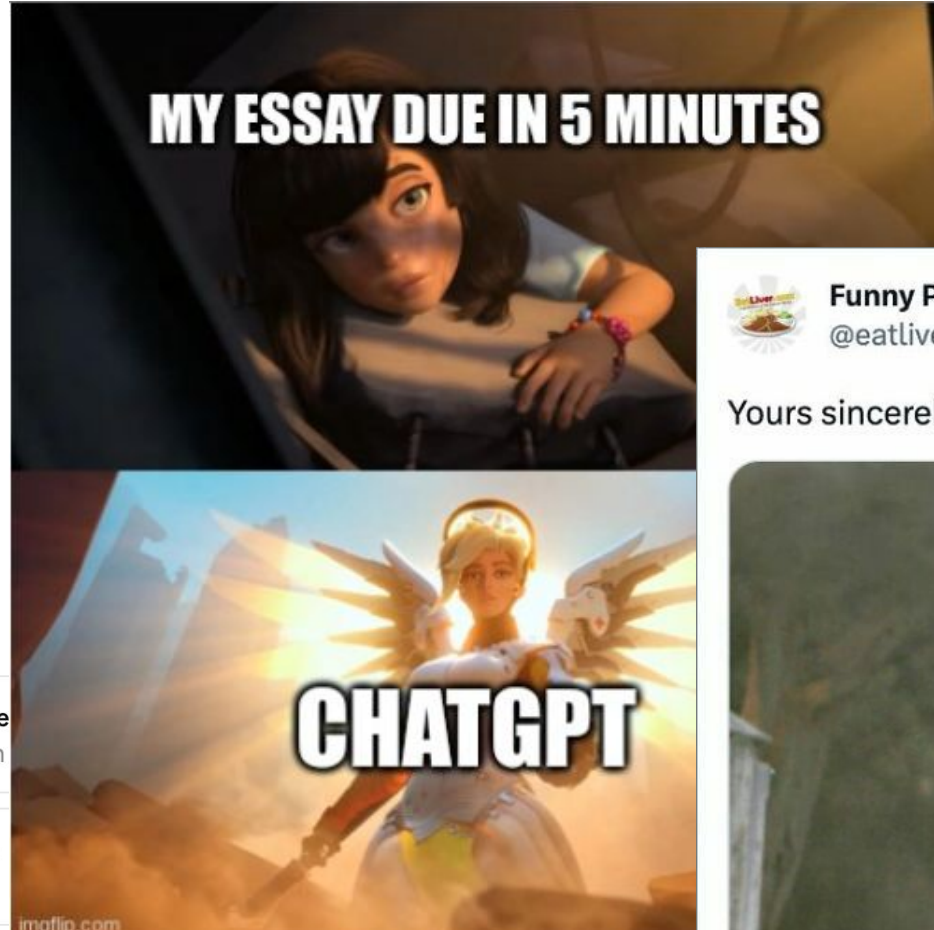


form content ide
new podcast on

Create a workout plan
for resistance training



Message ChatGPT...



Funny Pictures
@eatliver

Yours sincerely, ChatGPT.



2:19 AM · Jun 13, 2023

boredpanda.com

THE ChatGPT™

MY ESSAY DUE IN 5 MINUTES

I'm a programmer

Which programming
Languages You
Use

Chatgpt

CHATGPT

CHATGPT

ME LOOKING CLEVER
ON AN EMAIL

<https://locobuzz.com/blogs/chatgpt-memes/>



Funny Pictures

@eatliver

Yours sincerely, ChatGPT.



2:19 AM · Jun 13, 2023

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THE Chat



Peps

@PepsMccrea

BONUS → Best ChatGPT meme of 2022 🏆



Why are you so helpful?
What do you want in return?



As a language model trained by OpenAI, I
don't have wants or desires like a human
does.
But if you really want to help, you could give
me the exact location of John Connor.



9:56 PM · Jan 4, 2023

ny Pictures
atliver

cerely, ChatGPT.



in 13, 2023

boredpanda.com

The Stories of Little Bunny and Little Bear w/ Wise Owl





How can I help you today?

Brainstorm content ideas
for my new podcast on urban design

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in novels and in films

Create a workout plan
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Recommend a dish
to bring to a potluck

 Message ChatGPT...



ChatGPT can make mistakes. Consider checking important information.

The three-dimensional porous mesh structure of Cu-based metal-organic-framework - aramid cellulose separator enhances the electrochemical performance of lithium metal anode batteries

Manshu Zhang^{a,1}, Liming Wu^{a,1}, Tao Yang^b, Bing Zhu^a, Yangai Liu^{a,*}

^a Beijing Key Laboratory of Materials Utilization of Nonmetallic Minerals and Solid Wastes, National Laboratory of Mineral Materials, School of Material Technology, China University of Geosciences, Beijing 100083, China

^b College of Materials & Environmental Engineering, Hangzhou Dianzi University, Hangzhou 310036, China

ARTICLE INFO

Keywords:

Lithium metal battery
Lithium dendrites
CuMOF-ANFs separator

ABSTRACT

Lithium metal, due to its advantages of high theoretical capacity, low density potential, is used as a negative electrode material for batteries and brings great energy storage systems. However, the production of lithium metal dendrites poor safety, so lithium dendrites have been the biggest problem of lithium metal. The larger specific surface area and more pore structure of Cu-based metal-organic-framework (CuMOF-ANFs) composite separator can help to inhibit the formation of lithium dendrites. After 110 cycles at 1 mA/cm², the discharge capacity retention rate of the Li-Cu battery using the CuMOF-ANFs composite separator is about 96%. Li-Li batteries can continue to maintain low hysteresis for 2000 h at the same current density. The results show that CuMOF-ANFs composite membrane can inhibit the generation of lithium dendrites and improve the cycle stability and cycle life of the battery. The three-dimensional (3D) porous separator provides a new perspective for the practical application of lithium

1. Introduction

Certainly, here is a possible introduction for your topic: Lithium-metal batteries are promising candidates for high-energy-density rechargeable batteries due to their low electrode potentials and high theoretical capacities [1,2]. However, during the cycle, dendrites forming on the lithium metal anode can cause a short circuit, which can

chemical stability of the separator is equally important as it ensures that the separator remains intact and does not react or degrade in the presence of the electrolyte or other battery components. A chemically stable separator helps to prevent the formation of reactive species that can further promote dendrite growth. Researchers are actively exploring different materials and designs for separators to enhance their mechanical strength and chemical stability. These efforts aim to create separators that can effectively block dendrite formation, thereby improving the safety and performance of lithium-ion batteries. While there are several research directions to address the issue of dendrite formation, using a separator with high mechanical strength and chemical stability is an important approach to prevent dendrites from infiltrating the cathode and ensure safe operation of lithium metal batteries.

The three-dimensional porous mesh structure of Cu-based metal-organic-framework - aramid cellulose separator enhances the electrochemical performance of lithium metal anode batteries

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Certainly, here is a possible introduction for your topic: Lithium-metal batteries are promising candidates for high-energy-density rechargeable batteries due to their low electrode potentials and high theoretical capacities [1,2]. However, during the cycle, dendrites forming on the lithium metal anode can cause a short circuit, which can affect the safety and life of the battery [3-9]. Therefore, researchers are indeed focusing on various aspects such as negative electrode structure [10], electrolyte additives [11,12], SEI film construction [13,14], and collector modification [15] to inhibit the formation of lithium dendrites. However, using a separator with high mechanical strength and chemical stability is another promising approach to prevent dendrites from infiltrating the cathode. By incorporating a separator with high mechanical strength, it can act as a physical barrier to impede the growth of dendrites. This barrier can withstand the mechanical stress exerted by the dendrites during battery operation, preventing them from reaching the cathode and causing short circuits or other safety issues. Moreover,

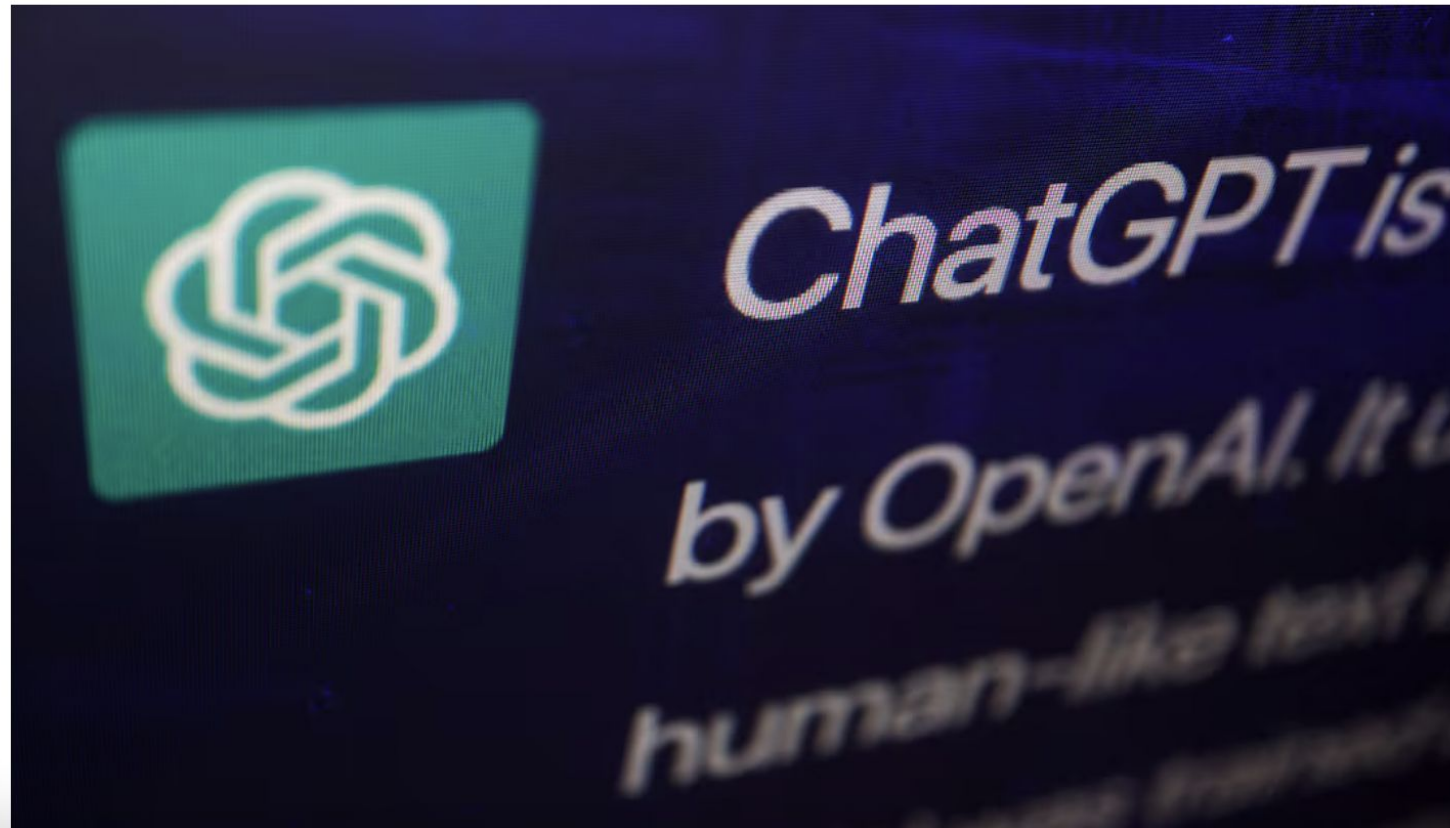
chemical stability of the separator is equally important as it ensures that the separator remains intact and does not react or degrade in the presence of the electrolyte or other battery components. A chemically stable separator helps to prevent the formation of reactive species that can further promote dendrite growth. Researchers are actively exploring different materials and designs for separators to enhance their mechanical strength and chemical stability. These efforts aim to create separators that can effectively block dendrite formation, thereby improving the safety and performance of lithium-ion batteries. While there are several research directions to address the issue of dendrite formation, using a separator with high mechanical strength and chemical stability is an important approach to prevent dendrites from infiltrating the cathode and ensure safe operation of lithium metal batteries.

Several types of separators currently used in research include nanoporous polymer separators [16], ceramic composite separators [17], nanofiber separators [18-20], and metal-organic skeleton (MOF) separators [21-24]. While these separators have shown some ability to inhibit the growth of lithium dendrites, they still have some drawbacks,

New York lawyers sanctioned for using fake ChatGPT cases in legal brief

By Sara Merken

June 26, 2023 4:28 AM EDT · Updated 2 years ago



GenAI offers a wide range of possible uses in education compared to past technologies



fix writing grammar

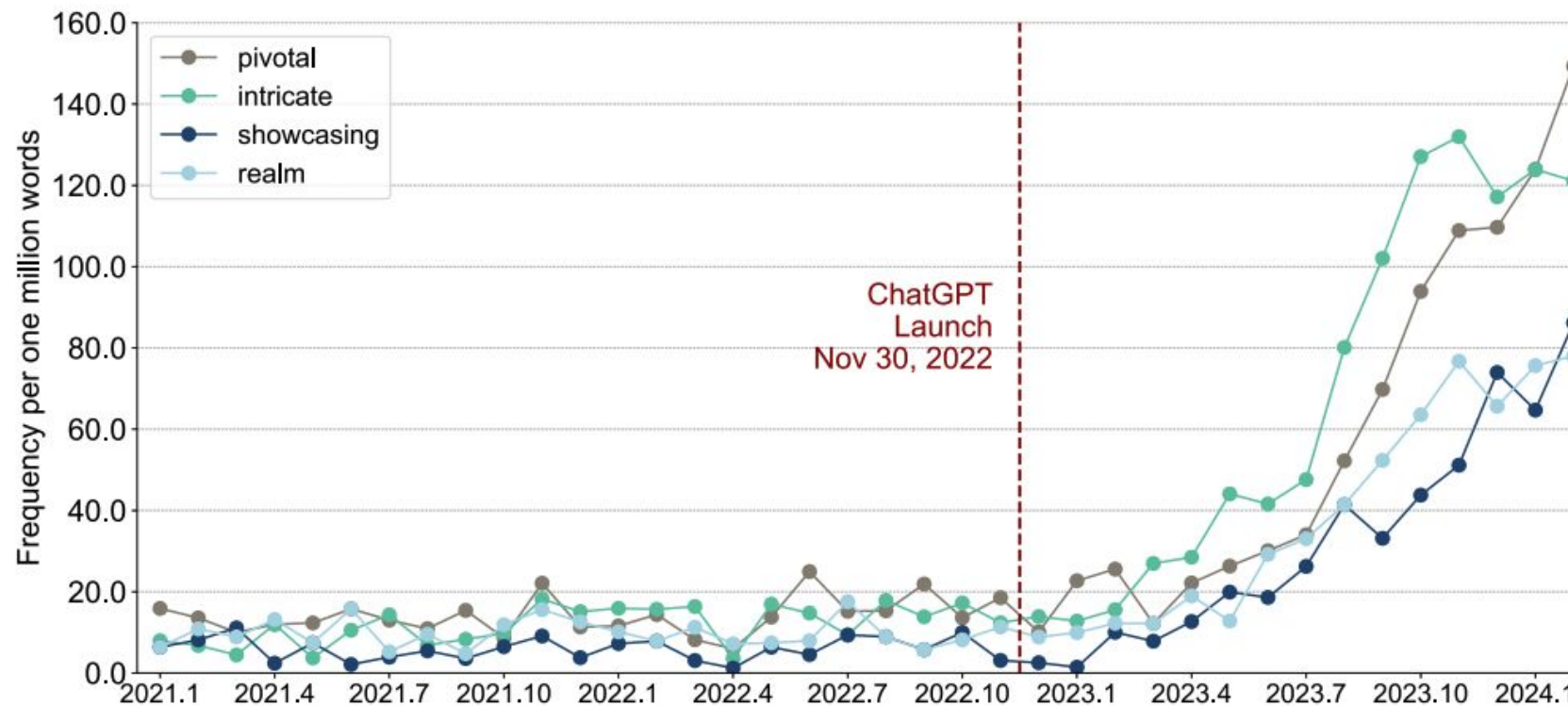


generate images



debug codes

GenAI has transformed the way people report new scientific findings



What is the implication of this observation?

Word Frequency Shift in arXiv Computer Science abstracts over 14 years (2010- 2024). The plot shows the frequency over time for the top 4 words most disproportionately used by LLM compared to humans, as measured by the log odds ratio. Source: <https://arxiv.org/pdf/2404.01268>

However, not all usages of generative AI benefit students' learning experiences and outcomes

RESEARCH-ARTICLE | OPEN ACCESS | 

The Value, Benefits, and Concerns of Generative AI-Powered Assistance in Writing

Authors:  [Zhuoyan Li](#),  [Chen Liang](#),  [Jing Peng](#),  [Ming Yin](#) | [Authors Info & Claims](#)

CHI '24: Proceedings of the CHI Conference on Human Factors in Computing Systems • Article No.: 1048, Pages 1 - 25
<https://doi.org/10.1145/3613904.3642625>

Published: 11 May 2024 [Publication History](#)



“Generative AI-powered assistance is found to offer benefits in **increasing people’s productivity and confidence** in writing. However, direct content generation assistance offered by AI also comes with risks, including **decreasing people’s sense of accountability and diversity** in writing.”

Discussion

What benefits and harms do you perceive in using generative AI tools (e.g., ChatGPT) in this class? Why?

Questions? Comments?

